

Pranav Reddy

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PROJECTS & EXPERIENCE

Quantum Codesign Lab

Niu Group, UCSB Undergraduate Researcher

June 2025 — present

- Developed feedback-based quantum algorithms (FQAs) for ground state Hamiltonian ground state preparation, applying Lyapunov based control to improve stability and convergence in hardware-constrained quantum systems
- Engineered large scale parallel simulation infrastructure, distributing workloads across 40+ CPU cores to accelerate performance-critical experimentation
- Built and optimized multi-layer adaptive quantum circuit pipelines, integrating real time feedback to inform system level design of hardware accelerated architectures

Pando.AI

Engineer Intern

<https://pando.ai>

July 2024–January 2025

- Built a real time retrieval-augmented automation pipeline with LLM-based inference to supply chain workflows
- Developed scalable alerting infrastructure and orchestration logic to improve response reliability and operational throughput in production enterprise environments

Real-Time Quantum Error Decoding (3-Qubit Bit-Flip Code)

Language: Python

January 2026–Current

- Simulated continuous noisy measurement signals from a 3-qubit error-detecting code, modeling hardware realistic error processes for fault-tolerant quantum systems
- Developed decoding pipelines to identify and correct bit-flip events in real time, benchmarking learned models against optimal Bayesian inference baselines
- Engineered scalable experimental infrastructure to evaluate performance across stochastic noise regimes, emphasizing latency, reliability, and hardware-constrained signal decoding
- Explored sequence based and graph based ML approaches (RNNs, GNNs) to improve error classification accuracy under compute constraints

EDUCATION

BS Electrical and Computer Engineering | University of California, Santa Barbara

September 2023 — Present

Pursuing electrical engineering major with an emphasis on photonics, high performance computing, computer architecture, ML/AI algorithm optimizations.

COURSE WORK

Intro to Photonics (ECE136A)

Probability and Stats (ECE139)

Computer Architecture (ECE152)

Quantum Photonics (ECE 136C)

Intro to Fields & Waves (ECE134)

Fiber Optic Systems (ECE 135)

Intro Solid State Devices (ECE132)

Intro to Optimizations (ECE 133)

Digital Circuit Design (ECE 10)

Problem Solving in C++ (CS 24)

Signal Processing (ECE 130A)

Waveguide Physics (ECE 144)

SKILLS & LEADERSHIP

Languages : C++, C, Python, Java, Arduino, LATEX, SystemVerilog, low-level Linux

Tools : Unix, Git/Github, LangChain, Tavily, OpenAI, Claude, Gemini, Quartus, Logisim, FPGA

Sigma Phi Epsilon - Vice President / Housing Manager / Siblinghood Chair

September 2023 - present